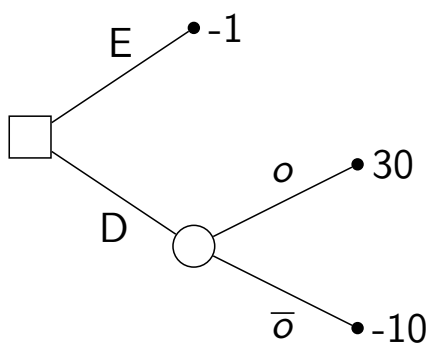


Lesson 7.1: The role of information

Information dynamics

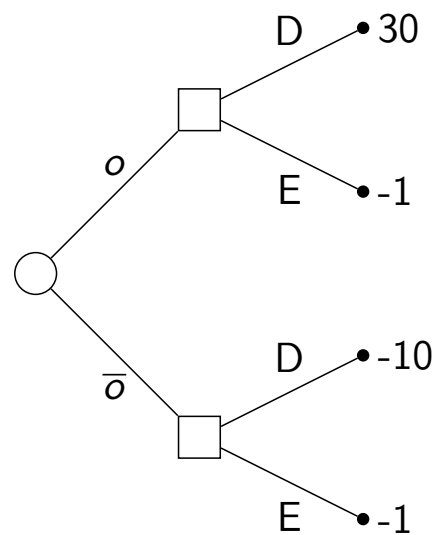
Victor Jauregui

Information



Alternatives:

- ▶ E
- ▶ D



Additional alternatives:

- ▶ if o, then E; otherwise D
- ▶ if o, then D; otherwise E

Oil prospecting: perfect information

- ▶ suppose even chance of oil being present; sonar test can determine if oil present
- ▶ effect of information (sonar):

	o	$\bar{o}$	EMV
E	-1	-1	-1.0
D	30	-10	10.0
ED	-1	-10	-5.5
DE	30	-1	14.5

- ▶ information improves outcome

strategy	EMV
D	10.0
DE	14.5

Value of information

- ▶ should you pay for information (\$3)?

strategy	EMV
D	10.0
DE	11.5

- ▶ how much should you pay?

strategy	EMV
D	10.0
DE	14.5

should be willing to pay up to $\$4.5 = 14.5 - 10.0$

- ▶ optimal strategy:

Pay for test; if oil present, then drill (D), otherwise, let lease expire (E)